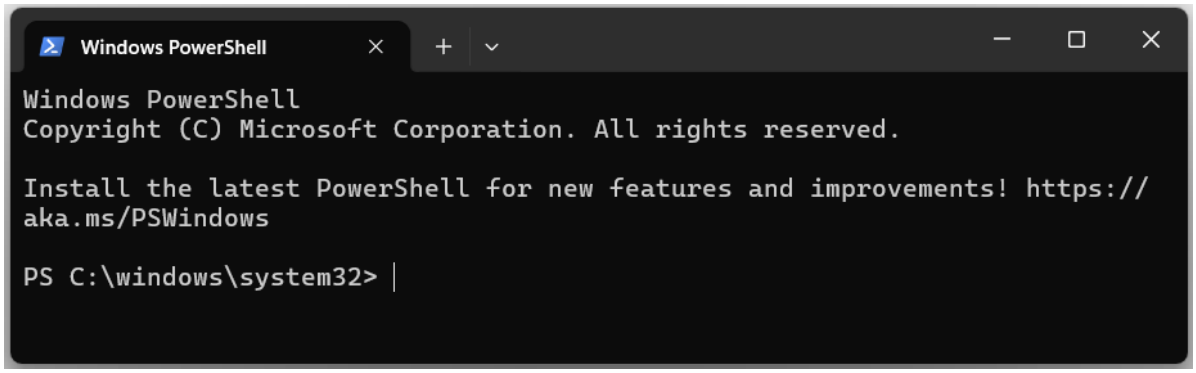
1. Installing Miniconda

Installing Miniconda (2 min)

1.1. Open terminal

Windows

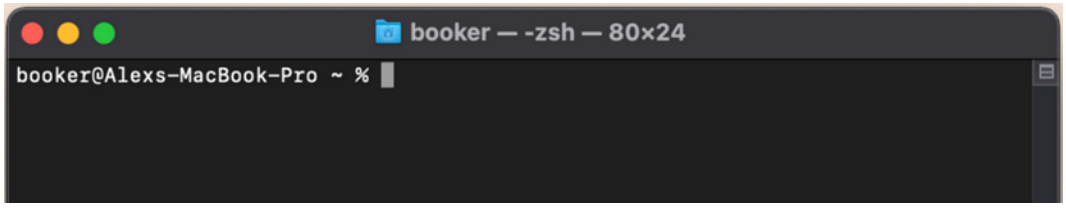
Search “Windows PowerShell” in the Windows Search Bar and open it.



Follow the instructions for “Windows PowerShell” in 1.2.

MacOs

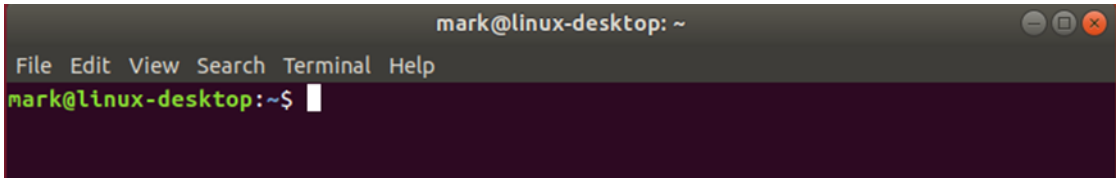
Type “Terminal” in the Search bar or look for it in Utilities and open. You should see a terminal like this:



Follow the instructions for “macOS” in 1.2.

Linux

Look for “Terminal” in the Activities item and open. Alternatively try Ctrl + Alt + T. You should see a terminal like this:



Follow the instructions for “Linux” in 1.2.

1.2. Download and Install Miniconda. Go to "https://www.anaconda.com/docs/getting-started/miniconda/install" and select your OS in the "Quickstart install instructions" tab:

Quickstart install instructions

These command line instructions will get you set up quickly with the latest Miniconda installer. Follow the steps for your system to download and install Miniconda, then follow the steps in **Verify your install** above to verify your Miniconda installation.

These quick install commands run a silent install. If you run a silent install, you are accepting Anaconda's Terms of Service (TOS) by default. Please make sure to review Anaconda's full TOS here before proceeding with silent installations.

Windows Command Prompt | Windows PowerShell | macOS | Linux

These commands work best with default PowerShell.

These three commands quickly and quietly download the latest 64-bit Windows installer, rename it to a shorter file name, perform a silent install, and then delete the installer:

```
Invoke-WebRequest -Uri "https://repo.anaconda.com/miniconda/Minico
Start-Process -FilePath ".\miniconda.exe" -ArgumentList "/S" -Wait
del .\miniconda.exe
```

* **VERY IMPORTANT!** Follow all the instructions that they provide in 3 steps, DO NOT stop after executing the commands from step 1. It is recommended to run line by line.

Potential error in Windows 11

During the Miniconda installation, if you get error messages like these:

```
Invoke-WebRequest : Access to the path 'C:\windows\system32\Miniconda3-latest-Windows-x86_64.exe' is denied.

At line:1 char:1

+ Invoke-WebRequest -Uri "https://repo.anaconda.com/miniconda/Miniconda ...

+ ~~~
```

You need to run this command on the PowerShell terminal:

```
cd $HOME
```

and then try again with the commands suggested in the miniconda webpage (step 1.2).

2. Visual Studio Code installation

2.1. Installing Visual Studio Code (3 min)

Download and install Visual Studio Code (VSC) using this link:

https://code.visualstudio.com/Download

2.2. Installing extensions of Visual Studio Code (3 min)

2.2.1. Open VSC

2.2.2. Go to the extensions panel (squares icon in the left sidebar) and search for all the next extensions.

- **Jupyter**
- **Prettier - Code formatter**
- **Python**
- **Python Indent**
- **Rainbow CSV**

3. Downloading Workshop Materials (when available)

3.1. Download the files needed for these lessons.

3.2. Create a folder called "ML_course" on your computer. **VERY IMPORTANT!** To avoid issues, do not create the folder in a OneDrive folder with cloud synchronization.

3.3. Move the downloaded materials to the new folder.

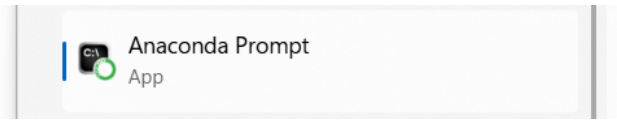
3.4. Unzip the file.

4. Virtual Environment Configuration

Python's virtual environment is a tool that creates an independent space to manage projects and its modules and versions. In other words, it creates various workspaces on one serve/computer

4.1. Open a new terminal:

4.1.A. Windows: after installing Miniconda (Step 1.2), you can now open an “Anaconda prompt”. Look for this tool in your applications an open it.



4.1.B. In macOS and Linux, open a terminal as you did in step 1.1.

4.2. Create a New Conda Environments (named cheminf & almos)

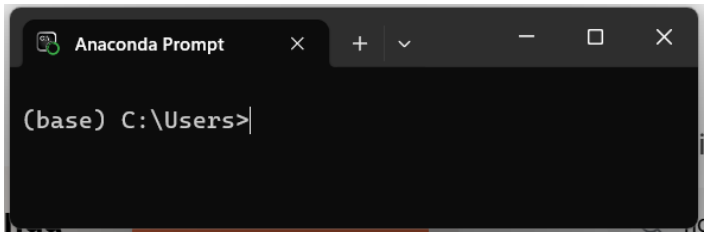
4.2.1. Go to the “ML_course/Prerequisites/Setup_and_testing” folder created in step 3. For this, you need to use the “cd” command in the terminal. For example, if your “ML_course/Prerequisites/Setup_and_testing” folder is in:

C:/Users/ML_course/Prerequisites/Setup_and_testing

You need to type the follow command in your terminal:

```
cd C:/users/ML_course/Prerequisites/Setup_and_testing
```

You can see in which folder your terminal is. For example, in the image below, the terminal is in the “C:/Users” folder.



4.2.2. Enter the two following commands separately. Note some installations may take a few minutes.

```
conda env create -f cheminf.yaml

conda env create -f almos.yaml
```

5. How to use Visual Studio Code and check the set up

5.1. cheminf environment

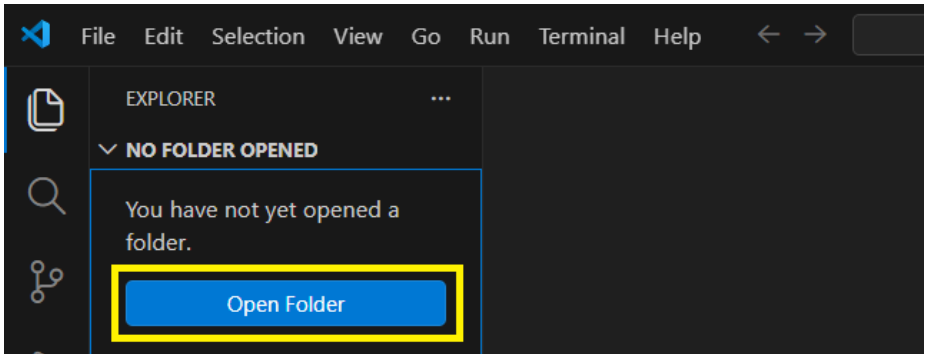
5.1.1. Open, using VSC, the “CHEMINF_course_setup.ipynb” Jupyter Notebook unzipped from the previous step (in the “ML_course/Prerequisites/Setup_and_testing” folder).

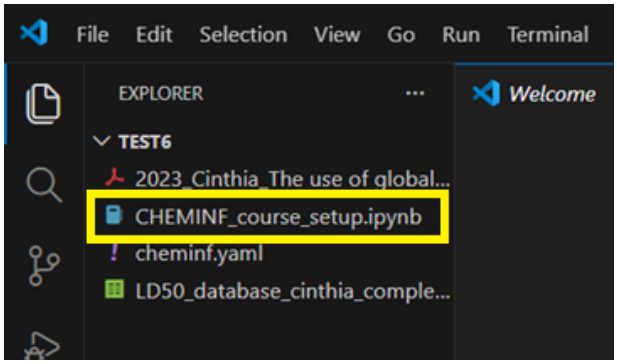
* (.ipynb means Interactive Python Notebook)

VERY IMPORTANT!

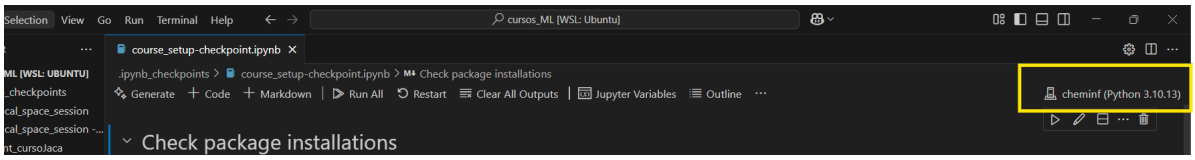
To open the notebook, do not double-click the file with your mouse. Instead, first open VSC and select “Open Folder”. You will then see the files in your folder listed on the left side of the screen. From there, select the notebook to open it.

If you don’t follow this process, VSC might not be able to detect the kernels from the points below.

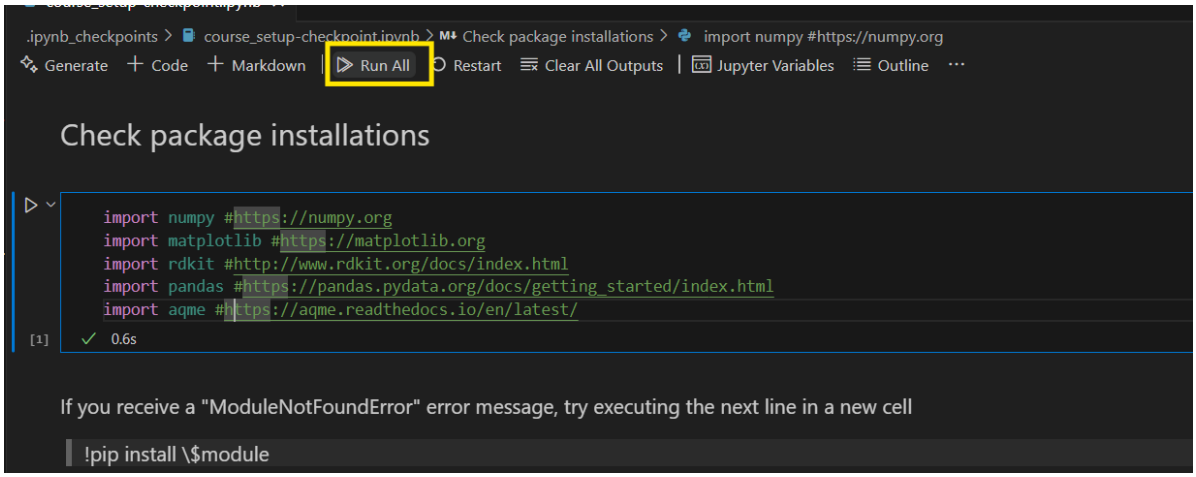




5.1.2. When VSC opens, select the cheminf kernel



5.1.3. Run the code cells by clicking “Run All” (image), placing the cursor on each cell and pressing the play button, or pressing *Shift* + *Enter*.



5.1.4. Check that each cell indicates that the library or program have been installed correctly.

5.1.5. Close the “CHEMINF_course_setup.ipynb” Jupyter Notebook and, then, close VSC.

<u>5.2. almos environment</u>

5.2.1. Open, using VSC, the “ALMOS_course_setup.ipynb” Jupyter Notebook unzipped from the previous step (in the “ML_course/Prerequisites/Setup_and_testing” folder).

5.2.2. When VSC opens, select the almos kernel (see step 5.2, but this time with “almos”)

5.2.3. Run the code cells by clicking “Run All” (image), placing the cursor on each cell and pressing the play button, or pressing *Shift* + *Enter (see step 5.3)*

5.2.4. Check that each cell indicates that the library or program have been installed correctly.

5.2.5. Close the “ALMOS_course_setup.ipynb” Jupyter Notebook and, then, close VSC.

6. Checking the CSV files

*A **CSV (Comma-Separated Values)** file is a simple text file used to store tabular data, where each line represents a row and values are separated by commas. It's commonly used for exchanging data between programs like spreadsheets and databases.*

Possible errors when working with CSV files in Windows:

Try to open a CSV file from the course folder, for example: test.csv (from set_up folder)

If when opening a CSV file everything is condensed into a single column, you can change this in your system's settings.

Moreover, in Excel to change the decimal separator in **File → Options → Advanced, and in the "editing options" section (first one), uncheck the "use system separators" and change the decimal separator to a period instead of a comma**. You can erase the thousand separator. Then click ok, and the settings should have changed so that the .csv files function correctly.

7. Testing easyROB

7.1. Portable Version (Windows)

Option A — Video Tutorial (Recommended)

If you prefer a visual step-by-step explanation, please watch the installation video using the hyperlink provided in this document.

How to install EasyROB

The video explains how to download the correct file, extract it, launch easyROB, and what to do if Windows shows a security warning.

Option B — Written Installation Instructions

For Windows users, installation is simple and takes only a few minutes.

7.1B.1.Go to the Releases page and scroll down to the "Assets" section.
https://github.com/MiguelMartzFdez/easyROB/releases

7.1B.2. Download the file called: **easyROB_win.zip**

7.1B.3. Once downloaded, right-click the ZIP file and select "Extract All...". Choose a simple location (for example, the Desktop).

7.1B.4. Open the extracted folder and double-click **launcher.vbs**. The program will start automatically

Fallback Method! (If the Portable Version Does Not Start)

If the portable version does not launch correctly (for example, nothing happens, the window closes immediately, or you see environment/Qt errors), you can run easyROB using the standard Terminal method.

Please follow the instructions described in Section 7.3.

If this situation occurs, please let us know and include:

- Your operating system and version

- A brief description of what happens
- Any error message shown
- The file **easyrob_process.log** generated in the folder

7.2. Portable Version (Linux)

7.2.1. Go to the Releases page and scroll down to the "Assets" section.
https://github.com/MiguelMartzFdez/easyROB/releases

7.2.2. Download the file called: **easyROB_linux.zip**

7.2.3. Once downloaded, right-click the ZIP file and select "Extract All...". Choose a simple location (for example, the Desktop).

7.2.4. **First-time setup only**: right-click inside the folder (in an empty area), select "Open in Terminal", and run:

```
chmod +x run_easyrob.sh
```

7.2.5. In the same Terminal window, run:

```
./run_easyrob.sh
```

Fallback Method! (If the Portable Version Does Not Start)

If the portable version does not launch correctly (for example, nothing happens, the window closes immediately, or you see environment/Qt errors), you can run easyROB using the standard Terminal method.

Please follow the instructions described in Section 7.3.

If this situation occurs, please let us know and include:

- Your operating system and version
- A brief description of what happens
- Any error message shown
- The file **easyrob_process.log** generated in the folder

7.3. Testing easyROB from Terminal (macOS / Fallback Method)

This method runs easyROB using a standard ROBERT installation (non-portable).

Steps

7.3.1. Open a Terminal.

Windows:

Open **Anaconda Prompt** (do not use the standard Windows Command Prompt).

macOS:
Press Cmd + Space, type Terminal, and press Enter.

Linux:
Press Ctrl + Alt + T or search for “Terminal” in your applications menu

7.3.2. Activate the conda environment where ROBERT is installed:

```
conda activate cheminf
```

7.3.3. Download **easyrob.py** from:

https://github.com/jvalegre/robert/blob/master/GUI_easyROB/easyrob.py

Click the 'Download raw file' button (top-right area of the page).

7.3.4. Navigate to the folder where you saved the file:

```
cd /path/to/your/folder
```

7.3.5. Launch easyROB by running:

```
python easyrob.py
```